

× PROOF-OF-REASONING

● MAINNET · LIVE

ERC-8004 REFERENCE IMPL

TuringVault

AI-powered RWA portfolio management infrastructure on Mantle — every allocation **survives adversarial challenge** before execution, anchored on chain.

ASSET CLASS

Tokenized Treasuries

USDT0 · mETH · USDY

CHAIN

Mantle Mainnet

5 verified contracts

VERIFICATION

3-model + ERC-8004

GLM-5 · Claude 4.6 · Gemini 3.5

WHAT MANTLE JUDGES SAID MATTERS

The hackathon brief lists **three defining features**. We built our entire stack around them.

"On-chain benchmarking of AI · ERC-8004 agent identity standard · radical transparency" — Mantle Turing Test 2026 brief

01 On-chain benchmarking of AI

Every cycle writes 4 attestation TXs to Mantle Mainnet (proposal · validation · decisionLog · reputation). 104+ decisions logged with full reasoning hashes. The hourly cron's public GitHub Actions workflow log gives judges a verifiable artefact, not a claim.

02 ERC-8004 agent identity standard

Reference implementation deployed on Mantle Mainnet — non-transferable identity NFT with auto-updating tokenURI, on-chain reputation accumulating from settled outcomes, validation registry for pre-action attestation. Ready to interop with Mantle's official Agent ID issuance.

03 Radical transparency

Public hourly cron · IPFS-pinned reasoning · live mascot reflecting real cycle freshness · /challenge page lets visitors probe the agent's gates · /discipline page shows post-execution proof per cycle · honesty-rule documented (.kiro/steering/no-lying-about-state.md) and enforced.

Live-stream ready: GitHub Actions workflow runs hourly, judges can drop into any past or current run.

WHY THIS MATTERS

AI agents allocate real capital. Nobody can verify why.

TODAY'S AUTONOMOUS AGENT

- Receives capital
- Black-box LLM produces decision
- Signs and broadcasts on-chain
- Reports PnL screenshot

No audit trail · No challenge · No accountability.

TURINGVAULT

- Receives capital
- **Three independent models challenge each other**
- **Every decision hashed → IPFS → Mantle**
- **Post-execution proof verification (Discipline Layer)**

Refusal-with-proof > profitable execution without one.

For RWA allocation specifically — DAO treasuries, DeFi-native funds, compliance-conscious operators — accountability is not a feature. It's the entire reason RWA assets exist.

OUR POSITIONING

Not a trading bot.

An accountable RWA portfolio manager.

RISK-ON LEG

mETH

Mantle native ETH staking — real yield

TREASURY ALLOCATION

USDT0

LayerZero-bridged Tether — Treasury-collateralised

RISK-OFF FLOOR

USDT

Stablecoin · pre-allocation buffer

DECISION QUALITY

65%+

Validator rejection rate · capital preservation

RWA EXPOSURE

74% NAV

Currently in tokenized Treasuries (USDT0)

PARSE STABILITY

100%

Structured-output success across 24h smoke

Compliance-conscious DAOs need an agent that can *prove its restraint*, not just its profits. Every blocked allocation has on-chain reasoning anchored to Mantle and pinned to IPFS.

HOW A DECISION FLOWS

Every allocation passes **three independent models** and three on-chain attestations. Then a fourth gate verifies the execution itself.

1. ANALYST**Z.ai GLM-5**

Proposes allocation from unified market context (CoinGecko, Nansen MCP, Hyperliquid funding, DeFiLlama TVL).

2. VALIDATOR**Anthropic Claude 4.6**

Adversarial gate. Default REJECT. Burden of proof on the proposer. R/R $\geq 1.5:1$ mandatory.

3. ARBITER**Google Gemini 3.5**

Tiebreaker on disagreement. 2-of-3 consensus required to execute.

```
Hourly GitHub Actions cron (public verifiable log)
|
v
unified market data → Analyst → Validator → Arbiter
|
v
4 on-chain attestations: proposal · validation · decisionLog · reputation
|
v
RWA Allocator (Path A LLM-driven · Path B deterministic idle-parking)
|
v
Merchant Moe LB v2.2 swap → Discipline Layer 3-gate post-exec verification
```

VERIFIABLE ON-CHAIN RIGHT NOW

Everything below is **verifiable on Mantlescan**.

102+

ON-CHAIN DECISIONS

65%

VALIDATOR BLOCK RATE

74%

NAV IN RWA

100%

PARSE SUCCESS (24H)

SMART CONTRACTS (SOURCIFY)

Identity
DecisionLog
ValidationRegistry
ReputationRegistry
Router

0x6f86...28bD
0x7bCd...cfbB5
0x6841...63b6
0xC781...6e1a
0x8187...7001

PUBLIC ARTIFACTS

First RWA swap (Mantlescan)
`0x0af2336...3e09de`
Hourly cron (GitHub Actions)
[Agent Cycle workflow](#) · [public log](#)
Reasoning blobs (IPFS via Pinata)
[CHALLENGE-*](#) · [PoR-*](#) · [AgentCard-v*](#)

Operator EOA `0xDC78...fb5a` · Mantle chain 5000 · all contracts verified

RADICAL TRANSPARENCY

We hold ourselves **to the same standard** we ask of the agent.

A project that promises Proof-of-Reasoning cannot afford fabricated metrics. Every claim on the dashboard is traceable to a contract read, a settled outcome, or a public file.

LIVE	AI consensus pipeline	102 on-chain TXs
LIVE	ERC-8004 identity + reputation	5 verified contracts
LIVE	Hourly cron via GitHub Actions	public workflow log
LIVE	Discipline Layer post-execution gate	3 gates fire each cycle
LIVE	RWA allocation to USDT0	first swap 0x0af2336...

Source-of-truth: `.kiro/steering/no-lying-about-state.md`

Infrastructure for **DAO-grade RWA allocation**, built natively on Mantle's primitives.

AI x RWA Track Path A scoring (40%): "Completeness of asset tokenization flow + innovation of technical approach". TuringVault is the rare submission that ships all three defining features end-to-end.

METH

Native ETH staking

Risk-on RWA leg with real on-chain yield, sourced from Mantle's own LST

USDT0

LayerZero omnichain Tether

\$1.1M Merchant Moe LB v2.2 pool — only liquid Treasury-collateralised target on Mantle

CRON

GitHub Actions hourly

~0.005 MNT / cycle gas → free-tier sustainable, public verifiable workflow log

SOURCIFY

5 contracts verified

Judge-readable source on Mantle Explorer, no opaque proxies

INNOVATION IN TECHNICAL APPROACH

- Three-model adversarial consensus with cross-provider diversity (Bedrock + Vertex AI)
- ERC-8004 reference implementation — auto-updating tokenURI per cycle
- Synrail-inspired Discipline Layer — TX proof + freshness + drift gates after every swap
- RWA Allocator dual-path: LLM-driven (Path A) + deterministic idle-parking (Path B)

WHO BUILT THIS AND WHAT'S NEXT

A solo developer. A repeatable pattern.

TEAM

USBVadik

Solo developer · ex-Synrail (autonomous-agent verification framework, also OSS)

github.com/USBVadik/TuringVault-Core

github.com/USBVadik/synrail

x.com/a_seven_life

STACK DISCIPLINE

→ Spec-first workflow (8 specs, >500 acceptance criteria)

→ 156/156 unit tests · CI on every push

→ Honesty rule enforced as steering doc

→ Every deploy verifiable on Mantlescan + GitHub Actions log

ROADMAP (POST-HACKATHON)

Q3 2026

Vault contract for public deposits · DAO treasury onboarding · multi-asset RWA basket

Q4 2026

Cross-agent reputation marketplace · ERC-8004 identity graph · per-decision IPFS streaming

2027

Open the Discipline Layer as an SDK · third-party agents stake reputation through TuringVault

LIVE DEMO

frontend-seven-beta-46.vercel.app

SOURCE

github.com/USBVadik/TuringVault-Core

CRON LOG

[Actions](#) / [Agent Cycle](#)

"The most valuable thing an AI RWA portfolio manager can do is **prove why it didn't allocate.**"